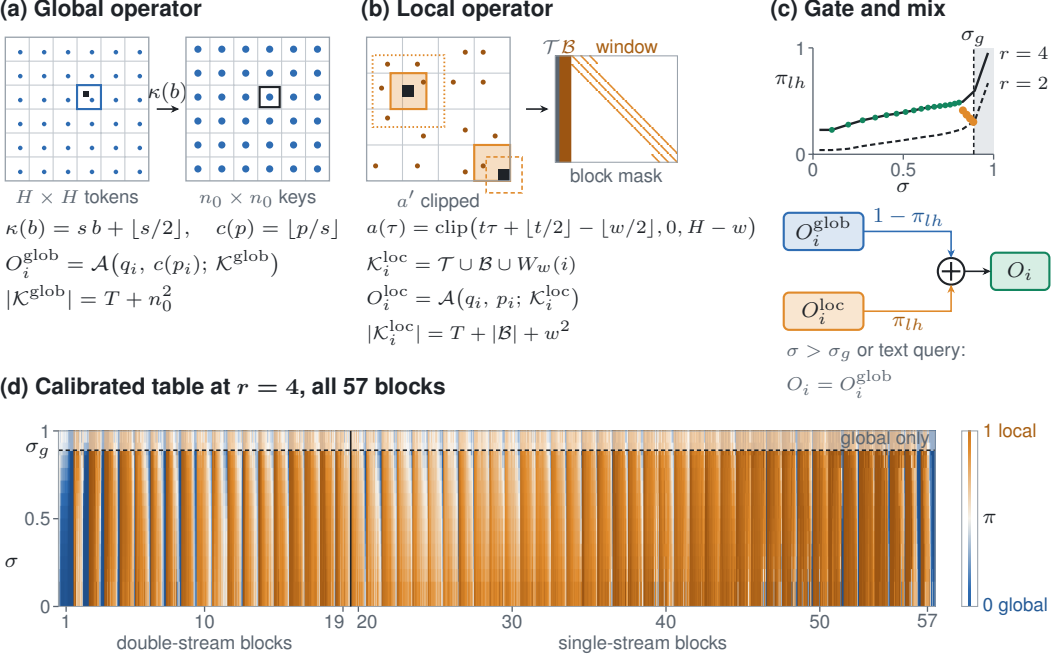

(a) Global operator
$\kappa(b)$
$H \times H$ tokens
$n_0 \times n_0$ keys
$\kappa(b) = s\,b + \lfloor s/2 \rfloor, \quad c(p) = \lfloor p/s \rfloor$
$O_i^{\text{glob}} = \mathcal{A}(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}})$
$|\mathcal{K}^{\text{glob}}| = T + n_0^2$
(b) Local operator
$\mathcal{T}\,\mathcal{B}$
window
block mask
a' clipped
$a(\tau) = \text{clip}(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w)$
$\mathcal{K}_i^{\text{loc}} = \mathcal{T} \cup \mathcal{B} \cup W_w(i)$
$O_i^{\text{loc}} = \mathcal{A}(q_i,\, p_i;\, \mathcal{K}_i^{\text{loc}})$
$|\mathcal{K}_i^{\text{loc}}| = T + |\mathcal{B}| + w^2$
(c) Gate and mix
σ_g
$r = 4$
$r = 2$
π_{lh}
σ
O_i^{glob}
$1 - \pi_{lh}$
O_i
O_i^{loc}
π_{lh}
$\sigma > \sigma_g$ or text query:
$O_i = O_i^{\text{glob}}$
(d) Calibrated table at $r = 4$, all 57 blocks
σ_g
σ
global only
1 local
π
0 global
double-stream blocks
single-stream blocks